

Analog Guitar Effect unit

Datasheet

Cybertron



Overview

Analog Guitar Effect Unit is an analog mixing console that does the preamplification and can add up to 3 independent effects. It has an equalizer, distortion, and tremolo effects that can be enabled and controlled individually. It is designed to do high-quality tone shaping with the most common and popular audio effects for a guitar.

Analog Guitar Effect unit

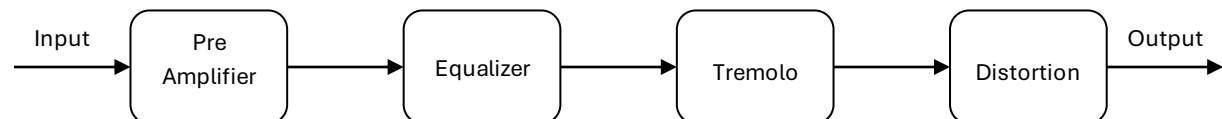
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Key Features

- 4-stage analog signal chain
- Pre-amp boost with Gain control
- 3-band active EQ (Low / Mid / High)
- Variable-Drive distortion stage
- Tremolo with independent Rate and Depth controls
- True bypass
- Low-noise analog design
- Standard 6.35 mm jack input and output

Signal Flow Diagram



Technical Specification

PARAMETER	MIN.	MAX.	UNIT
Supply Voltage	11	13.5	V (DC)
Supply Current	130	150	A (DC)
Input Voltage	0	500	mV
Input Current	0	0.01	mA
Output Voltage	0	4.5	V
Output Current	0	2.5	mA

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Bypass Type	True Bypass
Signal Path	100% analog
Input Impedance	40 k Ω
Output Impedance	207 Ω
Maximum Input	500 mV
Input port	6.35 mm
Output port	6.35 mm
Bandwidth	85 Hz – 15 kHz (–3 dB)

- Preamplifier

Maximum gain	9.58 dB
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- Equalizer

Low Range	80 Hz – 250 Hz
Mid Range	250 Hz – 1000 Hz
High Range	1 kHz – 15 kHz

- Tremolo

Tremolo Frequency	0.6Hz - 6.4Hz
Tremolo depth	80%

Mechanical Details

Dimensions	160 mm * 100 mm * 53 mm
Weight	200 g
Enclosure Material	Polylactic Acid (PLA)
PCB size	60 mm * 80 mm

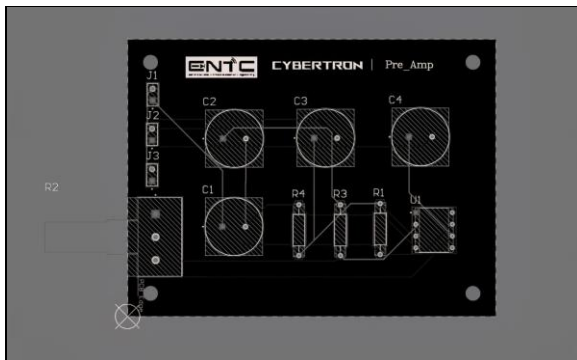
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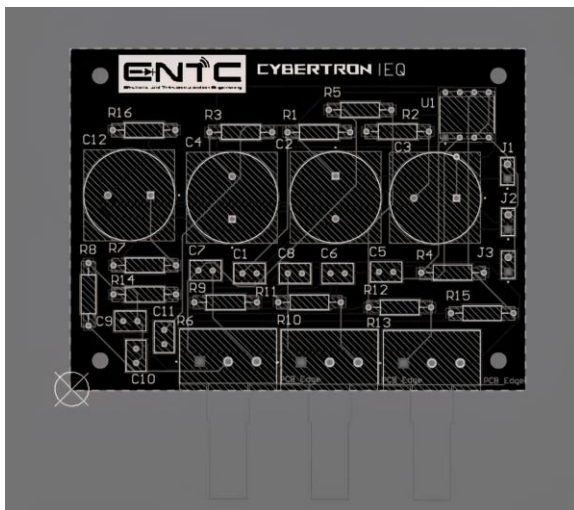
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PCB and Enclosure Overview

- Preamplifier PCB design



- Equalizer PCB design

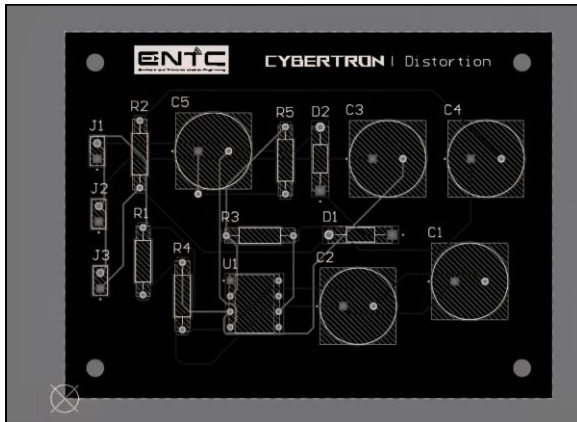


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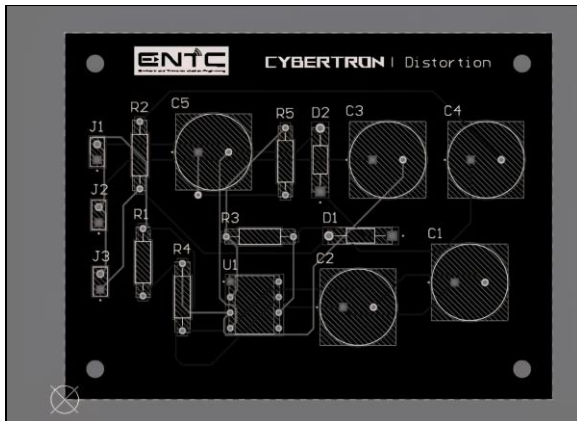
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- Distortion PCB Design



- Tremolo PCB Design

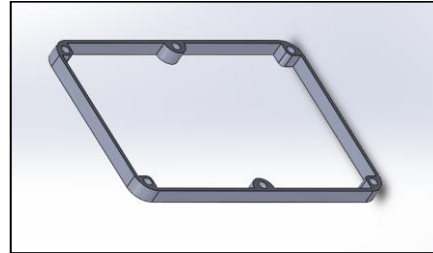
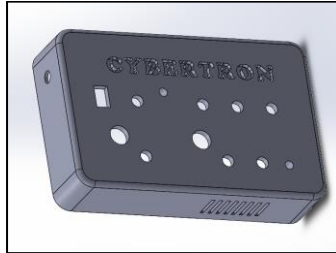
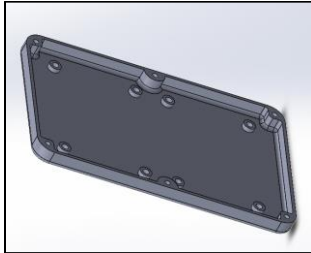


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- Enclosure Design



Environmental Parameters

PARAMETER	MIN.	MAX.	UNIT
Operating temperature	0	40	°C
Humidity	20	80	%

Safety Notes

- Keep away from water and moisture
- Do not open the enclosure
- Avoid long term exposure to direct sunlight